CROSS-BORDER ROLE OF BATTERY ENERGY STORAGE SYSTEMS (BESS) IN STABILIZING RENEWABLE POWER EXCHANGE BETWEEN GERMANY, POLAND, AND THE CZECH REPUBLIC

Key words: Battery Energy Storage Systems, Cross-Border Power Exchange, Renewable Energy Integration, Grid Stability, Central Europe

Danish ALI*
Ahtisham ARAIN*

The increasing penetration of renewable energy sources in Central Europe has introduced significant challenges related to power system stability and cross-border electricity exchange. Germany, Poland, and the Czech Republic are strongly interconnected through the European transmission network, where fluctuating wind and solar generation leads to congestion, loop flows, and renewable energy curtailment. [1] Battery Energy Storage Systems (BESS) have emerged as a key flexibility option to address these challenges by providing fast-response balancing, frequency support, and congestion relief. [2], [7] This paper investigates the role of BESS in supporting cross-border renewable energy integration in the DE–PL–CZ region. The study reviews regional renewable energy trends, existing interconnection infrastructure, and recent developments in large-scale battery storage deployment. Furthermore, technical, environmental, and economic impacts of BESS integration are discussed, highlighting its potential contribution to improving grid stability and strengthening regional energy cooperation.

1. INTRODUCTION

The European energy system is undergoing a rapid transformation driven by climate targets, energy security concerns, and the increasing deployment of renewable energy sources. Wind and solar power have experienced strong growth across Central Europe, particularly in Germany, while Poland and the Czech Republic are accelerating their transition away from fossil fuel–based electricity generation. [1], [9] Although renewable energy technologies significantly reduce greenhouse gas emissions, their variable and weather-dependent nature introduces new operational challenges for power systems.

* Brandenburgische Technische Universität, Cottbus-Senftenberg

In the interconnected electricity markets of Germany, Poland, and the Czech Republic, high renewable penetration has resulted in increased cross-border power flows, congestion of transmission lines, and frequency stability issues. Unplanned loop flows caused by excess wind generation in northern Germany frequently affect neighboring transmission systems, highlighting the need for additional system flexibility. [3], [5] Traditional solutions such as grid reinforcement and curtailment of renewable generation are often costly, time-consuming, or environmentally undesirable.

Battery Energy Storage Systems (BESS) offer a technically mature and rapidly deployable solution to address these challenges. [6], [7] Due to their fast response time and modular nature, batteries can provide multiple grid services including frequency regulation, peak shaving, congestion management, and renewable energy smoothing. This paper focuses on the role of BESS in stabilizing renewable power exchange between Germany, Poland, and the Czech Republic, emphasizing its importance for a resilient and cooperative Central European power system.

2. RENEWABLE ENERGY INTEGRATION AND CROSS-BORDER POWER EXCHANGE IN CENTRAL EUROPE

The power systems of Germany, Poland, and the Czech Republic are closely interconnected within the European Network of Transmission System Operators for Electricity (ENTSO-E). [1] Germany has the highest installed renewable energy capacity among the three countries [10], with large-scale wind farms in the north and increasing photo voltaic installations across the country. Poland and the Czech Republic, traditionally dependent on coal-fired power plants, are gradually expanding renewable generation while modernizing their transmission infrastructure.

Cross-border electricity exchange plays a crucial role in balancing supply and demand across the region. [9] However, high renewable generation in Germany often leads to unscheduled power flows through Polish and Czech transmission networks, causing congestion and operational stress. Phase-shifting transformers and market-based congestion management measures have been introduced to mitigate these effects, yet flexibility limitations remain. [3], [5] In this context, grid-scale battery storage systems are increasingly recognized as a complementary solution to support cross-border power flows and enhance system stability.

The three countries considered in this paper differ significantly in the share of electricity generated from renewable sources. Germany already exceeded 50% renewable electricity in 2023, whereas Poland and the Czech Republic remained at lower levels. This difference is relevant when assessing the present and future role of Battery Energy Storage Systems in supporting renewable integration and cross-border power exchange. [10]

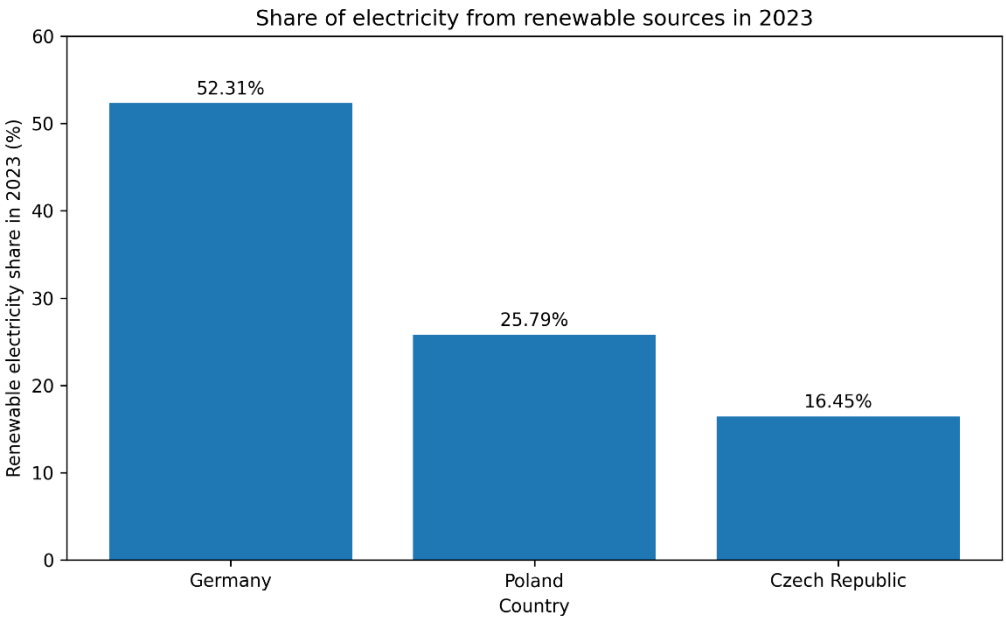

Fig. 1. Share of electricity from renewable sources in 2023 in Germany, Poland, and the Czech Republic
Data Source: Eurostat renewable electricity statistics, values accessed through CEIC country database pages [10].

As shown in Fig. 1, Germany's significantly higher renewable electricity share implies a greater need for flexibility measures such as storage and congestion management, while Poland and the Czech Republic are likely to face growing flexibility needs as renewable penetration increases further.

3. BATTERY ENERGY STORAGE SYSTEMS: TECHNOLOGY OVERVIEW

Battery Energy Storage Systems (BESS) have become one of the most widely deployed flexibility technologies in modern power systems. Their capability to store electrical energy and release it within milliseconds makes them particularly suitable for applications related to renewable energy integration and grid stabilization. Unlike conventional generation units, batteries are not limited by mechanical inertia and can respond almost instantaneously to grid disturbances.

Among the available battery technologies, lithium-ion batteries dominate current large-scale deployments due to their high energy density, declining costs, and proven

operational reliability. [6], [7] Other technologies such as sodium-sulfur batteries and flow batteries are also being investigated, especially for applications requiring longer discharge durations. However, lithium-ion systems currently represent the most mature solution for transmission and distribution grid applications in Central Europe.

From a technical perspective, BESS can be operated in different modes depending on grid requirements. These include frequency containment reserve, frequency restoration reserve, peak shaving, renewable energy smoothing, and congestion management. [2], [6] Modern power electronics-based converters enable batteries to provide both active and reactive power support, contributing to voltage regulation and improved power quality. This multifunctionality allows a single BESS installation to deliver multiple grid services, enhancing its economic viability.

In the context of renewable energy integration, batteries help mitigate short-term fluctuations caused by variable wind and solar generation. By absorbing excess generation during periods of high renewable output and supplying energy during deficits, BESS reduces the need for renewable curtailment and supports a more efficient utilization of existing generation assets. [7], [8]

The operation of a Battery Energy Storage System can be represented using a simplified power balance, where the change in stored energy depends on charging and discharging power over time.

$$E(t)=E(t-1)+\eta cPc(t)\Delta t-Pd(t)\Delta t/\eta d \tag{1}$$

where $E(t)$ is the stored energy at time t, $P_{(c)}$ and $P_{(d)}$ represent charging and discharging power, $\eta_{(c)}$ and $\eta_{(d)}$ denote charging and discharging efficiencies, and Δt is the time interval. This simplified model is commonly used to evaluate the short-term flexibility contribution of BESS in grid operation studies [6].

4. ROLE OF BESS IN CROSS-BORDER GRID STABILITY AND POWER EXCHANGE

Cross-border electricity exchange between Germany, Poland, and the Czech Republic is essential for maintaining regional supply security and efficient market operation. However, high shares of renewable energy generation, particularly wind power in Germany, have increased the frequency of congestion events and unscheduled power flows across national borders. [1], [3], [4] These loop flows can overload transmission corridors and require corrective actions by transmission system operators.

Battery Energy Storage Systems can play a significant role in addressing these challenges by providing localized flexibility near congested grid nodes. Strategically placed BESS installations can absorb surplus renewable energy before it enters constrained transmission corridors, thereby reducing cross-border congestion. [6], [8] During periods of high demand or reduced renewable generation, stored energy can be released to support neighboring systems and stabilize cross-border power flows.

Furthermore, BESS contributes to frequency stability across interconnected power systems. [6] Since frequency deviations propagate quickly across borders, fast-responding battery systems can support system-wide frequency control and reduce the reliance on conventional spinning reserves. This is particularly important in scenarios with declining conventional generation capacity due to coal phase-out policies in Poland and the Czech Republic.

In addition to technical benefits, BESS supports market-based cross-border integration by enhancing the effectiveness of day-ahead and intraday electricity markets. Improved flexibility reduces price volatility and enables better alignment between renewable generation and market schedules. As a result, battery storage contributes not only to technical stability but also to the economic efficiency of the Central European electricity market.

The role of Battery Energy Storage Systems in mitigating cross-border congestion and stabilizing power flows is illustrated conceptually in Fig. 2.

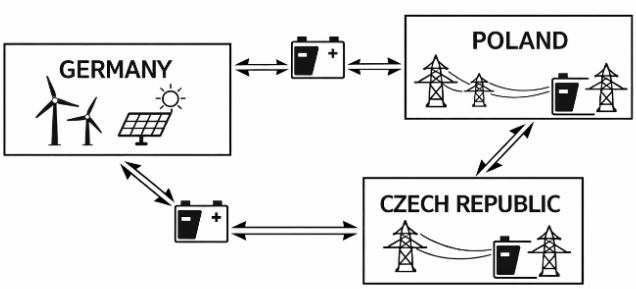

Fig. 2. Conceptual representation of cross-border power exchange and BESS integration in the Germany–Poland–Czech Republic region

5. COUNTRY-LEVEL DEVELOPMENTS AND CASE EXAMPLES

The role of Battery Energy Storage Systems in supporting renewable energy integration varies across Germany, Poland, and the Czech Republic due to differences in

generation mix, grid structure, and national energy policies. Nevertheless, all three countries increasingly recognize battery storage as a key element of future power system flexibility.

Table 1 summarizes recent developments in grid-scale battery storage deployment and primary applications in Germany, Poland, and the Czech Republic.

Table 1. Overview of BESS deployment and applications in the DE–PL–CZ region

Country	Typical BESS Size (MW)	Main Grid Services	Deployment Status
Germany	5–100	Frequency control, congestion relief	Commercial operation
Poland	1–50	Frequency regulation, peak shaving	Early deployment
Czech Republic	1–20	Grid stability pilots	Pilot projects

Table 1 was compiled by the authors based on publicly available system-development and annual-report documents from 50Hertz, PSE, and ČEPS [3]–[5].

5.1. GERMANY

Germany has the highest penetration of renewable energy among the three countries [10], with wind and solar power accounting for a significant share of electricity generation. Large-scale wind farms in northern Germany frequently generate surplus electricity, leading to congestion in north–south transmission corridors and increased cross-border power flows toward neighboring countries. To address these challenges, Germany has actively deployed grid-scale Battery Energy Storage Systems at both transmission and distribution levels.

Several large BESS installations are connected near renewable generation hubs and urban load centers, providing frequency control services and congestion relief. [3], [6] Batteries are increasingly participating in frequency containment reserve markets, replacing conventional power plants that are gradually being phased out. In addition, co-located battery systems at renewable power plants help smooth generation profiles and reduce curtailment during periods of excess supply. These developments demonstrate Germany's leading role in integrating storage technologies into market-based and grid-support frameworks.

5.2. POLAND

Poland's electricity system has historically relied on coal-fired power plants, which provided both energy and system stability. However, increasing renewable energy targets and aging thermal infrastructure are driving a transition toward cleaner generation. The rapid expansion of onshore wind and photovoltaic installations has increased the need for fast-response balancing resources, particularly as coal units are decommissioned or operated more flexibly.

Battery Energy Storage Systems are emerging in Poland as a complementary solution to support grid stability during this transition. Planned and recently commissioned battery projects focus on frequency regulation, peak load management, and renewable integration. [4], [9] From a cross-border perspective, BESS deployment helps mitigate unscheduled power flows originating from neighboring systems by providing localized flexibility. As Poland strengthens its transmission infrastructure and market integration with Germany and the Czech Republic, battery storage is expected to play an increasing role in maintaining operational security.

5.3. CZECH REPUBLIC

The Czech electricity system occupies a central position within the European transmission network and is strongly affected by cross-border power flows. Although renewable energy penetration is lower compared to Germany, the Czech grid experiences significant loop flows caused by renewable generation in surrounding countries. [5] These flows create operational challenges for transmission system operators, particularly during periods of high wind generation.

In response, the Czech Republic has focused on improving grid flexibility through a combination of network reinforcement, phase-shifting transformers, and pilot energy storage projects. Battery Energy Storage Systems are being evaluated for their potential to provide congestion management and frequency support services. While large-scale deployment is still in an early stage, pilot projects indicate that BESS can effectively support grid stability and reduce dependence on corrective cross-border measures. [5], [8]

6. DISCUSSION: TECHNICAL, ECONOMIC, AND ENVIRONMENTAL ASPECTS

The integration of Battery Energy Storage Systems into cross-border power systems presents several technical, economic, and environmental considerations. From a technical perspective, BESS offers fast-response flexibility that is particularly valuable in systems with high renewable energy penetration. In the Germany–Poland–Czech Republic region, batteries contribute to frequency stabilization, congestion management,

and improved utilization of renewable generation. However, their effectiveness depends strongly on optimal placement, sizing, and coordinated operation with existing grid infrastructure. [6], [8]

Economically, the deployment of large-scale BESS remains associated with high upfront investment costs, although these costs have declined significantly in recent years. [6], [7] Revenue streams such as frequency control services, energy arbitrage, and congestion management improve the economic feasibility of battery projects. In a cross-border context, coordinated market mechanisms are required to fully realize the benefits of storage, as national market designs and regulatory frameworks differ among the three countries. Without harmonized rules, the potential of BESS to support regional energy cooperation may remain underutilized.

From an environmental standpoint, Battery Energy Storage Systems support decarbonization by enabling higher integration of renewable energy and reducing reliance on fossil fuel–based balancing resources. By minimizing renewable energy curtailment and supporting coal phase-out strategies, batteries contribute indirectly to emission reductions. Nevertheless, environmental impacts related to battery manufacturing, raw material extraction, and end-of-life recycling must be considered. [7] Sustainable battery lifecycle management and recycling strategies will therefore play a crucial role in ensuring that storage deployment aligns with long-term environmental objectives.

7. CONCLUSION

The increasing integration of renewable energy sources in Central Europe has intensified the need for flexible and responsive grid solutions. In the interconnected power systems of Germany, Poland, and the Czech Republic, cross-border electricity exchange is both an opportunity and a challenge, particularly under conditions of high wind and solar generation. This paper has examined the role of Battery Energy Storage Systems in stabilizing renewable power exchange within this regional context.

The analysis shows that BESS can effectively support grid stability by providing fast frequency response, reducing congestion, and enabling better utilization of renewable energy resources. Country-level developments demonstrate that while Germany leads in large-scale deployment, Poland and the Czech Republic are increasingly recognizing the strategic importance of battery storage for their energy transitions. Despite remaining economic and regulatory challenges, coordinated deployment of BESS across borders offers significant technical and environmental benefits.

In conclusion, Battery Energy Storage Systems represent a key enabling technology for a resilient and cooperative Central European electricity system. [1], [2], [7] Their integration, supported by harmonized market frameworks and sustainable lifecycle strategies, will be essential for achieving long-term renewable energy and climate

targets in the Germany–Poland–Czech Republic region.

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